Palette Builder: Planning a Colour Palette in Conversation with a Language Model

A colour tool in which a language model asks about a painting before it proposes any colours, then comments each time the artist moves one

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Figure 1 — *The interface. The palette shown comes from a test against a stand-in server, written for the test; it was not produced by a model.*

ABSTRACT

Palette Builder is a colour tool for painters and illustrators. You describe the piece you are planning, a language model asks you about it, and only then does it propose a palette. The palette appears on a colour disc that you can rearrange yourself.

The model is asked to find out four things first: the medium, the subject, the format and audience, and the mood. The page will not show a palette until the medium, subject and mood have been recorded. After that the conversation carries on. When the artist drags a colour to a new place on the disc, the model is told what moved and comments on what the change does to the palette.

Every colour carries a tag saying who set it, and the page keeps a record of the whole session. **The record is the part I am most interested in, because it shows how a palette was arrived at and who made each decision.**

MOTIVATION

Choosing a palette is one of the first decisions in a painting, and one of the most personal. Colour tools usually treat it as geometry: pick a hue, choose a rule such as triadic or split-complementary, and the tool supplies the rest. A painting tutor works differently. They ask what the picture is about, what it is painted with and how it should feel, and their advice follows from the answers. I wanted to see what happens when a language model is asked to work in the second way.

Palette Builder sits between two of the modes of practice in my research. Building it was system co-creation: I wrote the instructions, the disc and the record-keeping, working with Claude as a coding assistant. Using it is artefact co-creation on a small scale, because each palette is made between the artist and the model. I wanted a tool that keeps the two contributions apart, so that they can be examined afterwards.

HOW A PALETTE IS MADE

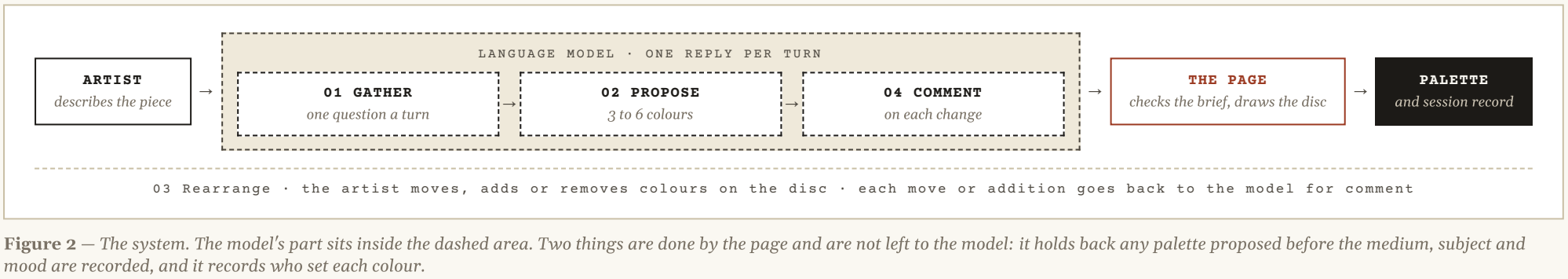


Figure 2 — *The system. The model's part sits inside the dashed area. Two things are done by the page and are not left to the model: it holds back any palette proposed before the medium, subject and mood are recorded, and it records who set each colour.*

The instructions ask the model to run the conversation in a particular way. Only the hold on early palettes is enforced in code. Everything else depends on the model following the instructions, and the session record shows when it does not.

01
GATHER

One question per turn, about the medium, the subject, the format and audience, and the mood. Each answer is filed in the brief.

Shown as four fields above the disc.

02
PROPOSE

Three to six colours, each with a name taken from the subject and a job in the painting, and the harmony described in plain words.

Held back until the medium, subject and mood are known.

03
REARRANGE

The artist drags any colour to a new hue or lightness, adds one or removes one. Dragging the base colour moves the whole palette together.

Tagged as moved or added by the artist.

04
COMMENT

The model is told what moved and from where, and says what the change does to temperature, value and harmony. It is asked not to move the colour back.

Removals are recorded but not sent for comment.

When I move a colour the model chose, whose palette is it?

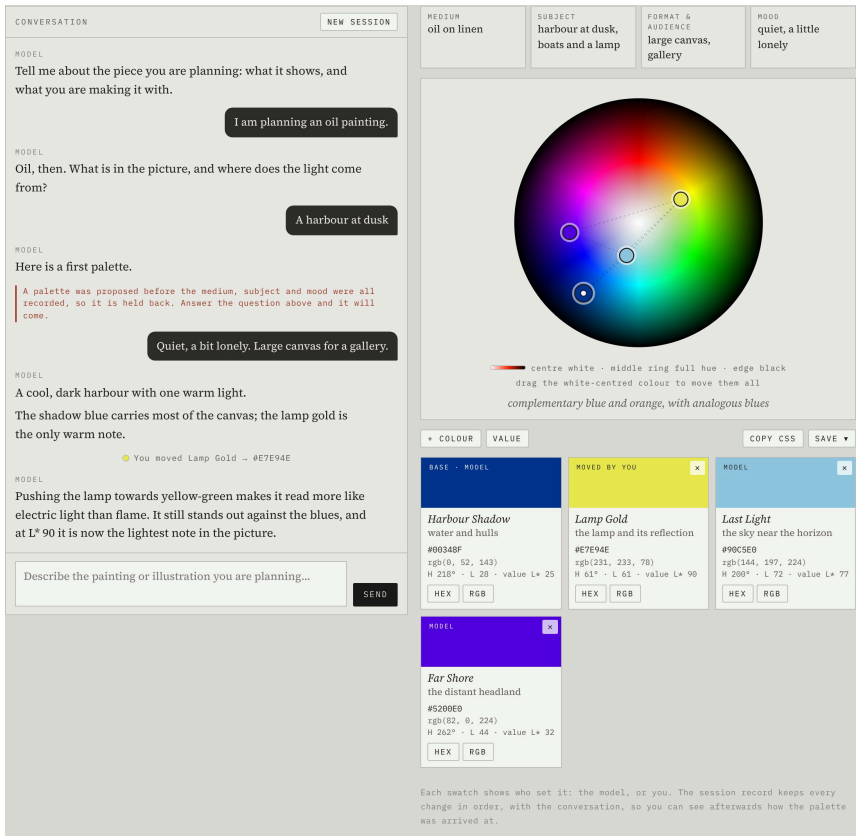


Figure 3 — *A session in progress, run against a stand-in server. The replies were written for the test. The first palette arrived before the mood was known and was held back. The gold was then moved by hand, and the model was asked about the change.*

THE COLOUR DISC

The disc puts hue around the circle and lightness along the radius: white at the centre, the full-strength hue on the middle ring and black at the edge. Painters think about value as much as hue, and this layout puts both under one drag.

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h = angle from the top, clockwise · hue, 0–359°
l = (1 − r / R) × 100 · lightness from the radius
s = min(2r / R, 1) × 100 · saturation follows
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Because saturation follows from position, the disc cannot show muted colours such as greys and earths exactly. The model is told this and asked to say when it matters. It is the main limit of the current design.

The VALUE button shows every colour as the grey of the same lightness (CIE L*), much as a painter squints to check that the darks and lights are where they should be. The page sits on a neutral grey because colours are judged against their surroundings (Albers, 1963).

WHO SET EACH COLOUR

Each colour is tagged MODEL, MOVED BY YOU or ADDED BY YOU. When the artist asks a question, the current palette and its tags are sent with it, so the model always works from the palette as it now stands.

The session record saves the conversation, every palette the model proposed, every change on the disc and every palette that was held back, in order and with times. It can be saved as a file at any point.

WHAT I WANT TO FIND OUT

No session with a real model has been recorded on this version yet, so this poster reports no findings. These are the questions the first sessions are designed to answer.

THE CONVERSATION

Do the questions change the palette? I want to compare palettes from a full conversation with palettes from a one-line request on the same subject, on both Claude and Qwen3.8.

Does the model keep to the order it is asked for, or does it try to propose colours early? The record shows every palette that was held back, so this can be counted.

THE ARTIST'S PART

How often, and how far, do artists move the colours they are given? The tags and the record answer this directly.

Does the model's comment on a change affect what the artist does next? I also want to run the same briefs with students and painting tutors, and compare their palettes with the model's.

WHO IS CHOOSING THE COLOURS?

In a finished palette some colours are tagged as the model's and some as mine, and the tags are accurate as far as they go. They do not show how much the model's questions shaped what I asked for, or how much my answers steered its proposals.

The session record keeps both sides of the conversation so that this can be traced, colour by colour. That tracing, more than the palette itself, is what connects the tool to my research.

THE OPPOSING VIEW

There is a simpler explanation of what the tool does. The model has read a great deal of writing about colour and mood, and it links loneliness to blue and joy to yellow because that is how people write about them. On this reading the questions are a form to fill in, and the comments would justify any move at all.

I take that view seriously. Comparing full conversations with one-line requests, and the model's palettes with tutors' advice on the same briefs, is meant to test it.

WHAT IT DOES NOW

- Runs inside Claude on the viewer's own account, or against a local model (Qwen3.8-27B-FP8 by default)
- Asks about the medium, subject, format and audience, and mood before proposing colours
- Holds back any palette proposed before the medium, subject and mood are known
- A hue-and-lightness disc with dragging, keyboard control and a value view
- Tags every colour with who set it, and comments on each change the artist makes
- Saves the palette as JSON or as a swatch card (SVG), and the whole session as a record
- A Jupyter notebook version for running briefs on a local or Hugging Face model

NEXT STEPS

- Record the first sessions, on Claude and on Qwen3.8
- Compare full conversations with one-line requests on the same briefs
- Add a saturation control, so that greys and earth colours can be shown properly
- Run the same briefs with students and painting tutors
- Read the session records alongside those from Sketch Lab

REFERENCES & STACK · ALBERS, J. (1963) *Interaction of Color* · ITTEN, J. (1961) *The Art of Color* · JOBLOVE, G. & GREENBERG, D. (1978) *Color spaces for computer graphics* · CLAUDE (ANTHROPIC) · QWEN3.8-27B-FP8 · HTML CANVAS 2D